EGCI 371
Computer Network

Layer 2: Data Link Layer

Layer 2 (Data Link)

- Explain **collisions** & **collision domains**
- See how **segmentation** (switch/VLAN) improves performance
- Compare **multiple access protocols** (ALOHA, CSMA/CD, CSMA/CA)
- Understand **LAN & MAC addressing** and **Ethernet frames**
- Trace **ARP** resolution end-to-end
  Say: “We’ll focus on practical Layer-2 behavior you can observe and debug.”

Collisions & Collision Domains

- **Collision**
- Simultaneous transmissions on a **shared medium** → corrupted frame
- Common with **hubs/half-duplex** Ethernet (legacy)
  Collision Domain
- Area where one device's transmission can collide with another
- **Hub/Repeater:** entire hub = 1 collision domain
- **Switch (full-duplex):** each port = separate collision domain → collisions effectively eliminated
  Say: "Modern switching removes collisions, but the concept matters"

Collision

- modern Ethernet switches isolate every link and run full-duplex , so there’s no longer a “shared wire” where frames can crash into each other.
- **Each port is its own collision domain (“micro-segmentation”)**
- A hub repeated bits to *all* ports → everyone shared one medium → collisions possible.
- A switch terminates the electrical/PHY on each port. Frames enter the switch, not the other hosts. That isolation alone prevents inter-host collisions.

Collision

Full-duplex links remove the collision condition

- With full-duplex, TX and RX use separate pairs/fibers; devices can transmit and receive simultaneously without interfering.
- Because collisions can’t happen on a full-duplex link, **CSMA/CD** (listen-before-talk, jam, backoff) is **disabled** and unnecessary.

MAC learning + unicast forwarding

- The switch learns (MAC → ingress port) from source MACs.
- When a frame arrives, it looks up the destination MAC and forwards only to the matching port (or drops if unknown, or floods in limited cases like unknown unicast/broadcast).
- Since only one egress link carries the frame, there’s no shared contention domain for it to collide in.

Collision

Store-and-forward (or cut-through) switching & buffering

- Store-and-forward: switch receives the whole frame, CRC-checks it, then forwards.
- Cut-through: begins forwarding after reading the destination MAC/EtherType (lower latency).
- Either way, per-port buffers/queues handle bursts; if the egress is busy, the frame waits in a queue—again, no “wire fight.”

Flow control instead of collisions

- If buffers fill, the switch/host can signal IEEE 802.3x PAUSE to momentarily throttle the sender (or rely on higher-layer congestion control/QoS).
- This avoids dropping frames due to congestion without ever relying on collision detection.

Collision

- **Switch fabric handles parallel conversations**
- Internally, a non-blocking (or low-blocking) switch fabric can move multiple frame streams at once between different port pairs—no mutual interference like a bus.

So why do you still see “collisions” counters sometimes?

- **Half-duplex links** (rare today) still use CSMA/CD and can show collisions.
- **Duplex mismatch** (one side full, other side half) causes **late collisions/CRC errors** on the half-duplex side and errors on the full-duplex side.
- **Wi-Fi** is different: shared RF medium uses **CSMA/CA**, so collision-*avoidance* is still a thing there—this doesn’t contradict switched Ethernet.

Segmentation (Reducing Contention & Scope)

Segmentation with Switches

- Hub → Switch: break big collision domain into per-port domains
  Broadcast Domain
- L2 switches **forward broadcasts** within the same VLAN
- **VLANs** segment broadcast domains logically (isolation, performance, security)
  Outcome
- Higher throughput, fewer faults impacting the whole LAN, easier troubleshooting
  Say: “Think in two scopes: collisions (per port) vs broadcasts (per VLAN).”

Multiple Access Protocols

Sharing the Medium

- **ALOHA / Slotted ALOHA:** simple, higher collision rate
- **CSMA:** listen before talk
- **CSMA/CD** (classic Ethernet, half-duplex): detect collision → jam → backoff
- **CSMA/CA** (Wi-Fi): avoid collision with backoff & ACKs (cannot detect collisions in the air)

Modern reality

- Switched **full-duplex Ethernet**: CSMA/CD largely irrelevant
- **Wi-Fi** still relies on CSMA/CA
  Say: "Ethernet ≈ wires & switches; Wi-Fi ≈ shared air—different MAC strategies."

Why Wi-Fi ≠ Switched Ethernet

- **Key idea:** *Shared air vs. dedicated wires*
- Wi-Fi shares a common RF medium (half-duplex)
- Can't "listen while talking" → can't detect collisions mid-transmit
- Therefore Wi-Fi uses **CSMA/CA** (Collision *Avoidance*), not CSMA/CD

Say: "Ethernet switches isolate links and run full-duplex; Wi-Fi must coordinate turn-taking over the air."

CSMA/CA in a Nutshell (DCF)

- Carrier Sense (CCA): listen first; transmit only if idle
- Random Backoff: choose a slot in the Contention Window (CW)
- Freeze/Resume: pause countdown if channel becomes busy
- Transmit → wait for ACK; no ACK ⇒ assume collision/loss
- Binary Exponential Backoff: enlarge CW after failures

Say: “ACK is how stations infer success—no ACK triggers a bigger backoff.”

Hidden Node & RTS/CTS

- Hidden node: A and C can't hear each other but both hear B → collisions at B
- RTS/CTS handshake:
  - Sender issues RTS (with Duration)
  - Receiver replies CTS
  - Neighbors set NAV (virtual carrier sense) and stay silent
- Use for large frames or noisy links (overhead trade-off)

Say: "RTS/CTS 'reserves' airtime so hidden neighbors don't step on each other."

Exposed Node & Spatial Reuse

- Exposed node: hears someone else’s frame and stays silent unnecessarily
- Modern Wi-Fi adds Spatial Reuse / BSS Coloring (Wi-Fi 6)
  - Mark frames by BSS color; far-away BSS can transmit concurrently

Say: “Not every transmission you hear will actually interfere—coloring helps reuse the air.”

Scheduling Helps: MU-MIMO & OFDMA (Wi-Fi 6/6E)

- MU-MIMO: AP serves multiple clients at once via spatial streams
- OFDMA: AP splits the channel into Resource Units (RUs) and triggers uplinks
- Result: fewer random contention rounds, higher efficiency at scale

Say: “The AP becomes a scheduler, shrinking the time we spend competing for airtime.”

Quick Compare & Troubleshooting

Ethernet (switched, full-duplex)

- No shared medium ⇒ no collisions; congestion handled by buffers, PAUSE, QoS

Wi-Fi (shared, half-duplex)

- Avoids collisions via CSMA/CA, ACK, RTS/CTS, EDCA; adds MU-MIMO/OFDMA/BSS coloring

Ops tips:

- Watch retries, missing ACKs, backoff, low SNR
- Prefer 5/6 GHz, right channel width; add APs to reduce cell size
- Enable WMM/EDCA, consider RTS/CTS for large frames/noisy links

Say: “Design for airtime efficiency: good SNR, right channels, and modern features enabled.”

LAN & MAC Address (Ethernet Frame Basics)

MAC Address (48-bit)

- 6 bytes: OUI (vendor) + device bits
- Types: Unicast, Multicast (I/G bit = 1), Broadcast = FF:FF:FF:FF:FF:FFEthernet Frame (simplified)
- Dest MAC | Src MAC | EtherType | Payload | FCS
- Typical MTU 1500 B (payload)
- Say: “At L2, delivery is by MAC to MAC, wrapped in a frame.”

ARP: IPv4→MAC Resolution

Purpose

- Map **IPv4 address to MAC** within a LAN
  Process
- Host sends **ARP Request (broadcast)**: "Who has 192.168.1.20?"
- Owner replies **ARP Reply (unicast)** with its MAC
- Both sides update **ARP cache** (temporary entries)

Notes

- **Gratuitous ARP**: self-announce / conflict detection
- **Security:** ARP spoofing/poisoning → MITM; mitigations: DAI, DHCP snooping, static ARP, NAC
  Say: "ARP is the glue between L3 addressing and L2 delivery."

Demo

Tryhackme

- Task 2: Nmap Live Host Discovery